Lesson 2.1 – 2.5 Outline & Review

[Lesson 2.1-2.2] Finding the **maximum/minimum of a quadratic function**
(find the vertex (h, k); then k = max/min and h = where the max/min occurs)

- Completing the square (changes standard form to vertex form)
- Averaging the zeros (factored form) ○
- Partial factoring (then averaging the x-coordinates of the two points)

[Lesson 2.3] Quadratic optimization problems
→ Remember to properly introduce your variables using "let" statements

- Number
- Area/Perimeter
- Revenue/Profit/Cost

[Lesson 2.4] Solving a quadratic equation for x ($0 = ax^2 + bx + c$)

Isolate for x	**Factoring**	**Quadratic formula**
• BEDMAS in reverse order • Use when given the form $0 = a(x - h)^2 + k$	• Factor to determine zeros • Use when factorable (when b2-4ac is a perfect square)	$x = \frac{-b \pm \sqrt{(b\text{^}2 - 4ac)}}{2a}$ • Use when not factorable

[Lesson 2.5] Quadratic equation problems
→ Remember to properly define your variables using "let" statements.
→Must have an equation that is either fully factored or in standard form before you solve.

- Number
- Area/Perimeter
- Revenue/Profit/Cost

Practice Questions

Lesson 2.1-2.2

1. Find the max/min of the function and state where it occurs using:
   a) Completing the square
      i. $y = 3x^2 - 4x + 2$ ii. $y = -2x^2 - \frac{1}{3}x - 1$ iii. $y = 2x^2 - 5x$
   b) Averaging the zeros
      i. $y = x^2 - 4x - 21$ ii. $y = 3x^2 - 5x + 2$ iii. $y = -3x^2 - 6x$
   c) Partial factoring
      i. $y = -\frac{1}{2}x^2 - 3x - 5$ ii. $y = 2x^2 - 5x - 8$ iii. $y = -5x^2 - 10x + 1$

Lesson 2.3

2. Two numbers have a difference of 10. If the sum of their squares is a minimum, find the two numbers.

3. Two integers have a difference of 82. Determine the minimum product of two numbers.

4. A schoolyard is fenced in on three sides. The total length of fencing is 120m. Find the maximum area of the schoolyard and its dimensions.

5. A rectangular garden against a wall is fenced on three sides and then divided into three equal sections. Find the dimensions that produce the maximum area of the garden if there is 600ft of fencing available.

6. The cost of a movie ticket is $12 and the expected attendance is 1000 people. If the ticket price is decreased by 25 cents, attendance will increase by 50 people. What ticket price will give the maximum revenue?

Lesson 2.4

7. Which method would be MOST efficient in each scenario?

a) Solve $4 = 25(x-2)^2 - 12$

b) Solve $0 = x^2 + 7x - 18$

c) Solve $0 = 30x^2 - 47x + 17$

d) The sum of two numbers is 16. What are the numbers if the sum of their squares is 200?

e) MovieBox sells 400 DVDs a month at $11 each. They determine that if the price of a DVD is increased by $1, they will sell 20 fewer DVDs a month. How many DVDs they must sell to have a revenue of $4600.

8. Explain why a quadratic equation must have the restriction $a \neq 0$ for $y = ax^2 + bx + c$

9. Define "solve." What are we doing when we "solve' an equation like $0 = 4x^2 + 5x - 2$?

10. Solve for x by

a) Isolating for x

i. $0 = -2(x+4)^2 + 2$ ii. $0 = 2(x - \frac{5}{2})^2$ Iii. $0 = 3(x-5)^2 + 4$

b) Factoring

i. $0 = -3x^2 - 6x$ ii. $0 = 3x^2 - 5x + 2$ iii. $2x^2 - 5x - 4 = 10 + x^2$

c) Quadratic Formula

i. $0 = -\frac{1}{2}x^2 - 3x - 5$ ii. $0 = 2x^2 - 5x - 8$ iii. $x^2 - 2x - 5 = 2x + 1$

Lesson 2.5

11. Two numbers have a difference of 5. Determine the two numbers if the sum of their squares is 193.

12. Tickets to a concert cost $100 and there are 2000 attendees. For every $5 increase in price, there are 50 fewer attendees. Determine the ticket price when the revenue is $216, 000.

13. An open-topped box is created by cutting out squares measuring 5cm by 5cm from a piece of square cardboard measuring x cm by x cm. If the capacity of the box is 500 cm^3, determine the size of the square cardboard.

Solutions:

1. Find the max/min of the function and state where it occurs using:

a) Completing the square

i. $y = 3x^2 - 4x + 2$

$= 3\left(x - \frac{2}{3}\right)^2 + \frac{2}{3}$

$\therefore$ *there is a* min *value of* $\frac{2}{3}$ *at* $x = \frac{2}{3}$

ii. $y = -2x^2 - \frac{1}{3}x - 1$

$y = -2\left(x^2 + \frac{1}{6}x + \left(\frac{1}{12}\right)^2 - \left(\frac{1}{12}\right)^2\right) - 1$

$y = -2\left(x + \frac{1}{12}\right)^2 - 2\left(-\frac{1}{144}\right) - 1$

$y = -2\left(x + \frac{1}{12}\right)^2 - \frac{71}{72}$

$\therefore$ *There is a* max *value of* $-\frac{71}{72}$ *at* $x = -\frac{1}{12}$.

iii. $y = 2x^2 - 5x$

$= 2\left(x - \frac{5}{4}\right)^2 - \frac{25}{8}$

$\therefore$ *there is a* min *value of* $-\frac{25}{8}$ *at* $x = \frac{5}{4}$

b) Averaging the zeros

i. $y = x^2 - 4x - 21$

$= (x - 7)(x + 3) \quad \therefore\ x_1 = 7\ \ x_2 = -3$

$x_v = 2$

$y_v = -25$

$\therefore$ *there is a* min *value of* -25 *at* $x = 2$

ii. $y = 3x^2 - 5x + 2$

$= (3x - 2)(x - 1) \quad \therefore\ x_1 = \frac{2}{3}\ \ x_2 = 1$

$x_v = \frac{5}{6}$

$y_v = -\frac{1}{12}$

$\therefore$ *there is a* min *value of* $-\frac{1}{12}$ *at* $x = \frac{5}{6}$

iii. $y = -3x^2 - 6x$

$= -3x(x + 2) \quad \therefore\ x_1 = 0\ \ x_2 = -2$

$x_v = 1$

$y_v = 3$

$\therefore$ *there is a* max *value of* 3 *at* $x = -1$

c) Partial factoring

i. $y = -\frac{1}{2}x^2 - 3x - 5$

$= -\frac{1}{2}x(x + 6) - 5 \quad \therefore\ x_1 = 0\ \ x_2 = -6$

$x_v = -3$

$y_v = -\frac{1}{2}$

$\therefore$ *there is a* max *value of* $\frac{1}{2}$ *at* $x = -3$

ii. $y = 2x^2 - 5x - 8$

$= 2x\left(x - \frac{5}{2}\right) - 8$

$\therefore\ x_1 = 0\ \ x_2 = \frac{5}{2}$

$x_v = \frac{5}{4}$

$y_v = -\frac{89}{8}$

$\therefore$ *there is a* min *value of* $-\frac{89}{8}$ *at* $x = \frac{5}{4}$

iii. $y = -5x^2 - 10x + 1$

$= -5x(x + 2) + 1 \quad \therefore\ x_1 = 0\ \ x_2 = -2$

$x_v = \frac{0-2}{2} = -1$

$y_v = 6$

$\therefore$ *there is a* max *value of* 6 *at* $x = -1$

2. The two numbers are 5 and -5.
3. The minimum product of two integers is -1681.
4. The maximum area is 1800m^2. The two equal sides are 30m and the remaining length is 60m.
5. The dimensions of 75m by 300m will maximize area.
6. The ticket price of $8.50 will give the maximum revenue.

7. a) isolating x b) factoring c) quadratic formula d) factoring e) quadratic formula
8. If a = 0, then y = bx + c which is the equation of a line, so we must have $a \neq 0$ so that we have the necessary ax^2 that makes an equation quadratic.
9. "Solve" means to find the solutions, or values, that will make the LS equal to the RS.
10. a- i) x = -5, -3 ii) $\frac{5}{2}$ iii) no sol b- i) x = -2, 0 ii) x = 2/3, 1 iii) x = -2, 7 c- i) no sol ii) $\frac{5 \pm \sqrt{89}}{4}$ iii) $2 \pm \sqrt{10}$
11. 7 and 12 or -7 and -12
12. $120 or $180
13. 20 cm by 20 cm